

Roll Number:_____

Indian Institute of Technology Madras
Department of Management Studies
MS 6015: Multivariate Statistical Methods for Business
Sample Quiz 1

1. This quiz has a total 7 questions worth 20 marks.
2. The duration of the quiz is 50 minutes.
3. Read all questions carefully.
4. Please write your answers in the space provided
5. *All the best.*

Section 1: Multiple Choice

1. Random samples of size 49 are taken from an infinite population whose mean is 300 and standard deviation is 21. The mean and standard error of the sample mean, respectively, are:
 - (a) 300 and 21
 - (b) 300 and 3
 - (c) 300 and 0.43
 - (d) None of these choices.
2. Suppose the ages of students in your program follow a positively skewed distribution with mean of 24 years and a standard deviation of 4 years. If we randomly sampled 100 students, which of the following statements about the sampling distribution of the sample mean age is NOT true?
 - (a) The mean of the sampling distribution of sample mean is equal to 24 years.
 - (b) The standard deviation of the sampling distribution of sample mean is equal to 4 years.
 - (c) The shape of the sampling distribution of sample mean is approximately normal.
 - (d) All of these choices are true.

3. The MBA program at DOMS enrolls students directly from undergraduate studies and others after having obtained some work experience. The students with work experience claim that they do better than their peers because of their experience in the business world. However, the students right out of school claim they do better than their peers because they have more recent academic experience.

The faculty at DOMS is not interested in which group does better, but simply in determining whether the two groups' performance is the same or not. To test this, the program took a random sample of students' grades (from a database). Let μ_W denote the population mean GPA for the students with work experience and let μ_N denote the population mean GPA for students with no work experience. Let $\bar{x}_W$ and $\bar{x}_N$ represent the corresponding sample means.

You analyze the data from the random sample of students' grades with a two-sample, two-sided t -test (which may or may not be correct), obtaining the following results:

Statistic	Value
Difference ($\bar{x}_W - \bar{x}_N$)	0.211 667
Std. Error of Difference	0.104 050
t-Ratio	2.034 271
Degrees of Freedom	55
Upper 95% CL	0.420 188
Lower 95% CL	0.003 145
$P(T > t)$ (Two-tailed)	0.0468
$P(T > t)$ (Right-tailed)	0.0234
$P(T < t)$ (Left-tailed)	0.9766
Confidence Level	0.95

What conclusion do you reach from the above output?

- (a) Reject the null hypothesis and conclude that the mean GPAs are different.
 - (b) Accept the null hypothesis and conclude that the mean GPAs are different.
 - (c) Reject the null hypothesis and conclude that the mean GPAs are the same.
 - (d) Accept the null hypothesis and conclude that the mean GPAs are the same.
4. What is the interpretation of the "Difference" value in the table?
- (a) The average GPA of those in the sample without work experience is 0.212 points higher than those with work experience.
 - (b) The average GPA of those in the sample with work experience is 0.212 points higher than those without work experience.
 - (c) Can't determine without looking at the original data.

Section 2: Short answer-

5. An investor studied the percentage rates of return of three different gold funds. Random samples of percentage rates of return for four periods were taken from each fund. The results appear in the table below and the overall mean is 8.4

	Fund 1	Fund 2	Fund 3
	12	4	9
	15	8	3
	13	6	5
	14	5	7
	17	4	4
means	14.2	5.4	5.6

The square of deviations of each value from the overall mean is given below

	Fund 1	Fund 2	Fund 3
	12.96	19.36	0.36
	43.56	0.16	29.16
	21.16	5.76	11.56
	31.36	11.56	1.96
	73.96	19.36	19.36
SUM	183	56.2	62.4

- (a) state the null and alternate hypothesis
 (b) Complete the Anova Table

Source of variation	SS	df	MS	<i>F</i>
Treatments				
Error	49.2			
Total				

6. Nutrient intake was measured for a random sample of 402 women aged 25-50 years. One of the questions of interest is whether women meet the nutritional intake guidelines. The table below shows the recommended daily intake and the sample means for iron and protein:

Variable	Recommended intake	Sample mean
Iron	15 mg	11 mg
Protein	60 g	66 g

The sample variance-covariance matrix S is $\begin{bmatrix} 30 & 100 \\ 100 & 1000 \end{bmatrix}$

Infer whether there is significant evidence at 5% level whether on the average the women meet the recommended guidelines

- (a) State the null and alternate hypothesis both in words and mathematical expression
- (b) Compute the test statistic
- (c) What is the critical value
- (d) Compare the test statistic with the critical value and state your conclusion.
7. Thirty patients participated in a clinical trial of a new weight-loss formula A. Among them, 10 patients were assigned to receive formula A, and independently sampled 10 patients received the old formula B. After 14 days, A-patients reported the average weight loss of 3 lbs, and B-patients reported the average weight loss of 2 lbs. After 28 days, A-patients reported the average weight loss of 7 lbs, and B-patients reported the average weight loss of 4 lbs. The sample covariance matrices between the 14-day loss and the 28-day loss were $\begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$ in the A-group and $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ in the B-group. Assuming equality of population covariance matrices and Bivariate Normal distribution of responses within each group, explain how you would test statistically if there is a difference between the expected effects of the two formulas. In particular, state the hypothesis and compute the test statistic.
8. Observations on two responses are collected for three treatments. The observation vectors $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ are

Treatment 1: $\begin{bmatrix} 6 \\ 7 \end{bmatrix}, \begin{bmatrix} 5 \\ 9 \end{bmatrix}, \begin{bmatrix} 8 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 \\ 9 \end{bmatrix}, \begin{bmatrix} 7 \\ 9 \end{bmatrix}$

Treatment 2: $\begin{bmatrix} 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Treatment 3: $\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

The break up the observations into mean, treatment, and residual for each of the variable is given below:

	Obs	Overall Mean	Trt effect	Error
Treatment 1	6	4	2	0
	5	4	2	-1
	8	4	2	2
	4	4	2	-2
	7	4	2	1
Treatment 2	3	4	-2	1
	1	4	-2	-1
	2	4	-2	0
Treatment 3	2	4	-1	-1
	5	4	-1	2
	3	4	-1	0
	2	4	-1	-1
SS	246	192	36	18

	Obs	Overall Mean	Trt effect	Error
	7	5	3	-1
	9	5	3	1
	6	5	3	-2
	9	5	3	1
	9	5	3	1
	3	5	-1	-1
	6	5	-1	2
	3	5	-1	1
	3	5	-3	1
	1	5	-3	-1
	1	5	-3	-1
	3	5	-3	1
	402	300	84	18

The Sum of squares of cross products of observations is 275, Mean is 240, Treatment is 48 and of residual is -13.

- a Construct the corresponding arrays for each variable.
- b Using the information in Part a, construct the one-way MANOVA table.

c Evaluate Wilks' lambda.